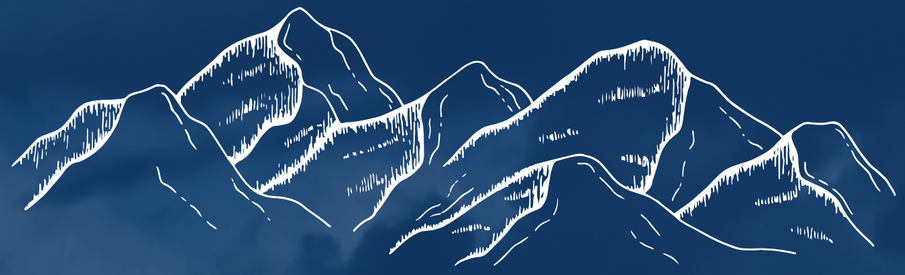
GROUP 6



EXPLAINABLE AI-BASED INTERFACE SYSTEM FOR WEATHER FORECASTING MODEL

OUTLINE

1. Introduction & Research Objectives
2. Research Methodology
3. Analysis of XAI Methods
4. Comparison & Discussion
5. Application & Case Study
6. Limitations & Future Research Directions
7. Conclusion Questions

Introduction & Objective

Problem:

Lack of explainability in black-box deep learning models for rainfall forecasting

Objectives:

1. Identify forecasters' needs.
2. Choose suitable XAI methods.
3. Evaluate explanations qualitatively and quantitatively.

Goal:

Design a user-centered XAI interface

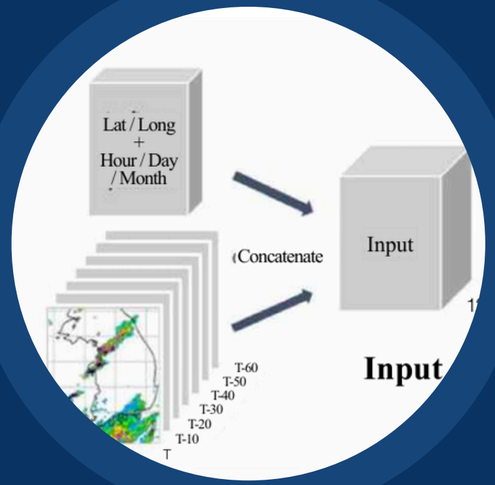
RESEARCH MEDTHODOLOGY

Task: Classify short-term rainfall into 3 levels.

- **Model:** UNet2 (extended DeepRaNE).
- **Input:** Radar images, geo-info, timestamps.
- **Output:** Rainfall intensity classification.
- **Data:** Korean radar data (2020).
- **Weather scenarios:** 5 classified by SOM.

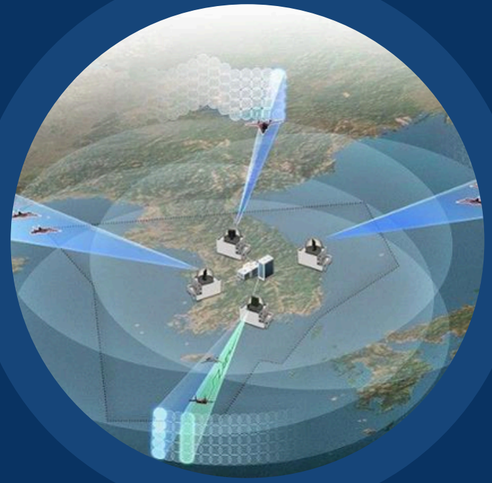
ANALYSIS OF XAI METHODS

UNET2 MODEL



UNet2 model for 1–6 hour precipitation intensity classification.

RADAR DATA



2020 Korean radar data classified into 5 weather scenarios.

USER SURVEY

16 online sessions, 3 expert consultations.

XAI METHODS

Select XAI methods aligned with each explanation requirement.

EXPLANATION 1: MODEL PERFORMANCE BY RAINFALL TYPES

Mapped Method

→ Rainfall-type classifier Performance Diagram

What it does

- Classifies input data into rainfall types (e.g., frontal, convective).
- Evaluates model performance per type using F1, POD, CSI.

Why it matters

- Reveals biases in model behavior across different weather patterns.
- Supports forecast validation and trust in specific conditions.

EXPLANATION 2: OUTPUT REASONING

Mapped Method:

→ Feature Attribution (e.g., Integrated Gradients, Saliency Maps)

What it does:

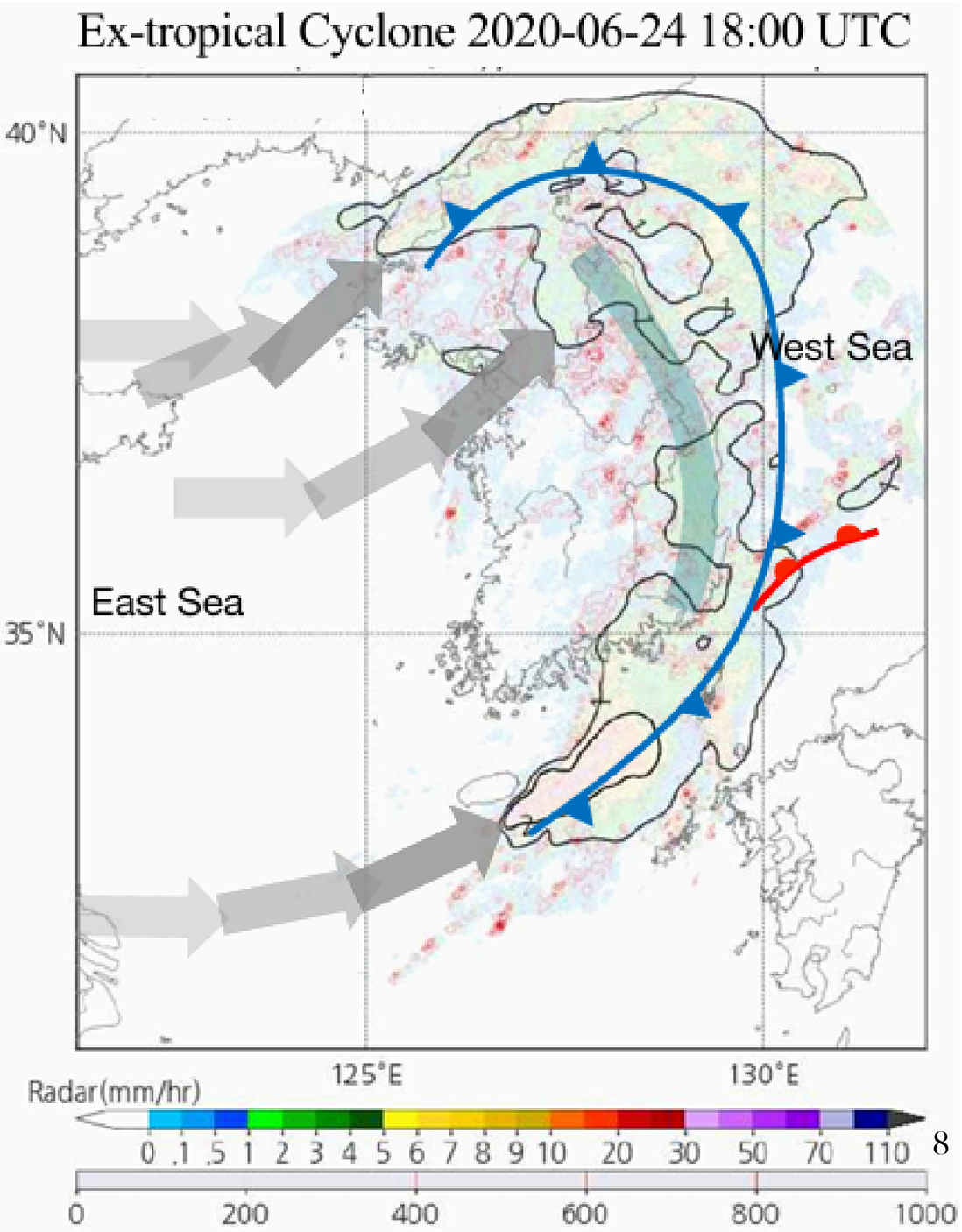
- Highlights which parts of the radar input influence predictions.
- Visualizes formation or decay of convective systems.

Evaluation:

- Quantitative: Completeness scores (via perturbation tests)
- Qualitative: Domain expert interpretation of real-world cases

Why it matters:

- Enables forecasters to understand the reasoning process of the model.



EXPLANATION 3: CONFIDENCE CALIBRATION

Mapped Method:

→ Reliability Diagram + Expected Calibration Error (ECE)

What it does:

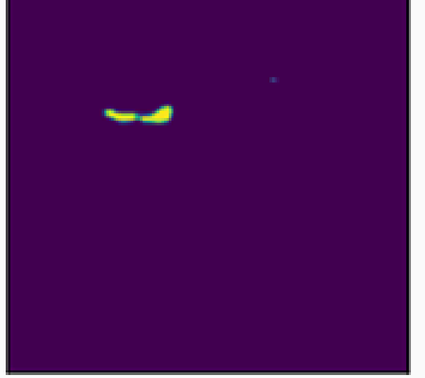
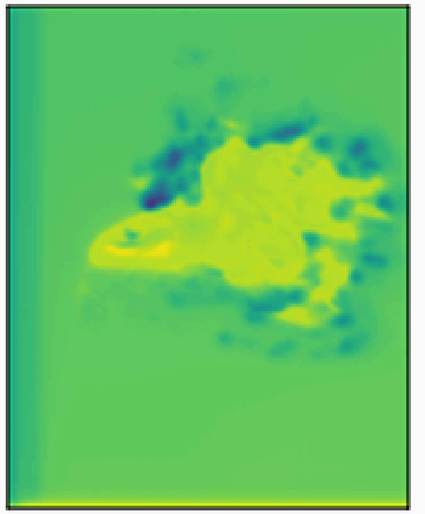
- Measures the alignment between predicted probability and actual outcomes.
- Visualizes calibration across different regions and lead times.

Evaluation:

- ECE score (lower is better)
- Reliability diagrams (closer to y = x line is ideal)

Why it matters:

- Provides trust level of each forecast
- Helps avoid over-reliance on overconfident predictions

Softmax (Confidence %)	
Logit/T	
Distribution of softmax values X: logit/T Y: softmax	

COMPARISION AND DISCUSSION

COMPARISION AND DISCUSSION

Performance Diagram: stable, intuitive, high usability
Integrated Gradients: high fidelity, low user accessibility
Calibration: reliable probabilities, no label change

Users favor intuitive visuals over pure fidelity

APPLICATION AND CASE STUDY

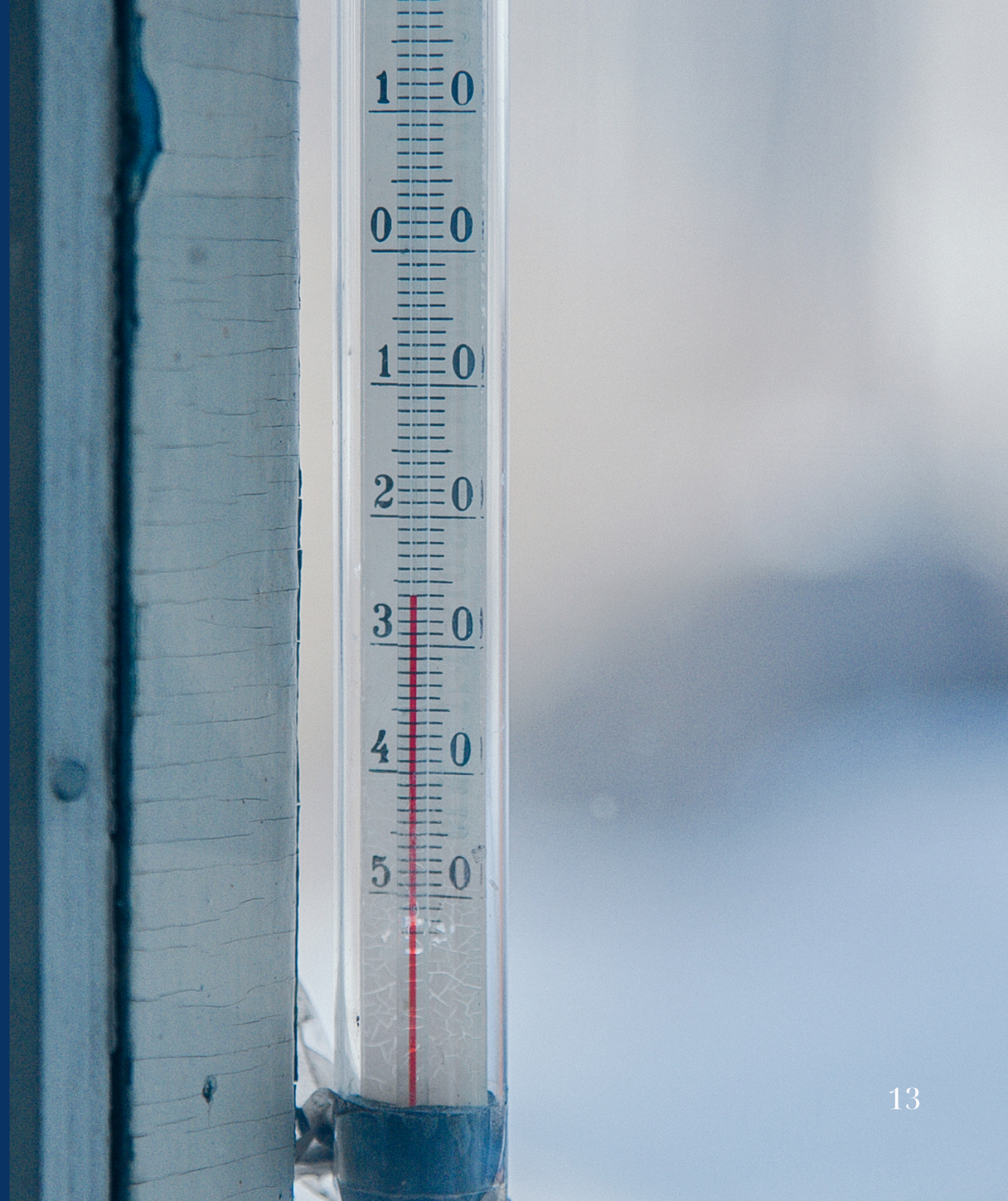
APPLICATION AND CASE STUDY

Implemented a prototype UI system.

Features:

- Interactive radar map with clickable grid points.
- Toggle between probability, attribution, and confidence views.
- Time-step navigation

Case study with real 2020 radar data demonstrated usability and effectiveness.



LIMITATIONS & FUTURE RESEARCH DIRECTIONS

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Limitations

- Small sample size, limited to 2020 radar data
- Lack of multi-modal data (satellite, NWP)

Future Work

- Expand surveys across regions and organizations
- Develop multi-dimensional explanations and interactive UI

CONCLUSION & QUESTIONS

THANK YOU!